JAGAT GURU NANAK DEV PUNJAB STATE OPEN UNIVERSITY, PATIALA

DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS

M.Sc. (Computer Science) — Semester-IV

APPLICATION FOR APPROVAL OF SEMINAR TOPIC

Course Code: MSCS-4-04 | Course: Seminar (M.Sc. Computer Science, Semester-IV)

To:
The Course Coordinator,
Department of Computer Science & Applications,
Jagat Guru Nanak Dev Punjab State Open University (JGND PSOU),
C/28, The Lower Mall, Patiala - 147001, Punjab.

---

1. Student Particulars

Name of the Student: `Sumit Singh`

Roll Number / Enrollment No.: `PSOU/MSCS/2026/042`

Program: Master of Science in Computer Science — M.Sc. (CS)

Semester / Academic Cycle: Semester-IV (Final Semester) / `July 2024 - 2026 Cycle`

Study Centre Code / Name: `JGND PSOU Regional Centre, Patiala`

Mobile Number / Email: `[Student Contact & Email]`

---

2. Proposed Seminar Particulars

Domain / Area of Research: Distributed Systems / Advanced Web Architectures / Data Consistency

Proposed Title of Seminar:
"Architectural Patterns for Offline-First Distributed Web Applications: Balancing Zero-Latency User Experience with Eventual Consistency"

Target Presentation Duration: 15–20 Minutes (with additional 5 Minutes for Discussion & Questions)

Expected Length of Report: 10–15 A4 Pages (3,000–4,500 Words)

---

3. Abstract / Significance of the Proposed Seminar Topic

In high-reliability enterprise web applications, the traditional "thin-client / thick-server" architecture introduces single points of failure, latency spikes, and complete operational failure during network disruptions. The emerging **"Local-First" and "Offline-First" distributed paradigm** shifts the primary operational data store directly into the client's browser runtime (via IndexedDB and local transactional storage engines), allowing instantaneous local state mutations ($< 1\text{ms}$) while decoupling remote database writes via asynchronous background synchronization queues.

This seminar will explore:

The theoretical foundations of local-first software, CAP theorem trade-offs, and distributed consensus models.

Conflict resolution mechanisms, including Conflict-Free Replicated Data Types (CRDTs), vector clocks, and idempotency key queues.

Modern browser-based transactional engines (IndexedDB, Dexie.js) paired with event-driven backend workers.

Real-world case study implementation and empirical performance benchmarks demonstrating how offline-first architectures achieve high fault tolerance and zero-latency user experiences in mission-critical software.

---

4. Preliminary References & Literature

Kleppmann, M., Wiggins, A., van Hardenberg, M., & McGranaghan, M. (2019). *"Local-first software: you own your data, in spite of the cloud."* Proceedings of the ACM on Human-Computer Interaction (PACMHCI), Vol. 3, Issue OOPSLA.

Shapiro, N., Preguiça, N., Baquero, C., & Zawirski, M. (2011). *"Conflict-Free Replicated Data Types."* International Symposium on Stabilization, Safety, and Security of Distributed Systems, Springer.

Brewer, E. (2012). *"CAP twelve years later: How the 'rules' have changed."* Computer, IEEE, 45(2), 23-29.

---

5. Undertaking by the Candidate

I hereby request approval for the above seminar topic. I undertake to submit the formal Seminar Report (10–15 A4 pages) within 30 days of topic approval and deliver the oral presentation and defense as scheduled by the university.

Date: `15/05/2026`
Place: Patiala / `Patiala`

Signature of Candidate: _____

---

6. For Official Use Only (Course Coordinator / Seminar In-charge)

Seminar Topic Status: [ ] Approved / [ ] Modification Required / [ ] Rejected

Seminar Supervisor / Evaluator: _____

Scheduled Date of Presentation: `15/05/2026`

Remarks:

Signature of Course Coordinator: _____
Date: `15/05/2026`